- The operational details of how an application gets to use the DLT for its intended purpose: verification, validation, and immutable recording for enabling trust
- The consensus algorithm (at a high level) for the integrity of the DLT

TRANSACTIONS, BLOCKS, AND CHAIN OF BLOCKS

Applications initiate transactions and the execution of smart contract code. A simple cryptocurrency transfer between accounts, for example, generates a “send” transaction. The transactions generated are broadcast through the blockchain network and then gathered and recorded in the distributed immutable ledger. Listing 1.1 shows an example pseudocode for function calls for initiating two types of transactions. Tx1 is for the transfer of cryptocurrency. Tx2 is an application-specific transfer of ownership of an asset from one owner to another, probably to fulfill the sale of an asset. You can also observe the use of the rule `onlyByOwner` for the `transferOwnership` function, which means that only the owner of the account can execute that function. Such rules are necessary for the autonomous systems that blockchain controls. In chapters 3-5, you’ll learn how to code rules like these.

Listing 1.1 Pseudocode for two functions initiating transactions

```
/Tx1: */ web3.eth.sendTransaction(fromAccount, toAccount, value);    ❶
/Tx2: */ transferOwnership(newOwner);                                ❷
```